

UNIT – I

Data mining: Introduction, Data mining – on what kind of data, data mining functionalities – what kind of patterns to be mined, Classification of data mining systems, data mining task primitives, integration of a data mining systems with a database or data warehouse systems, major issues in data mining.

Data preprocessing: Descriptive data summarization, data cleaning, data integration and transformation, data reduction, data discretization and concept hierarchy generation.

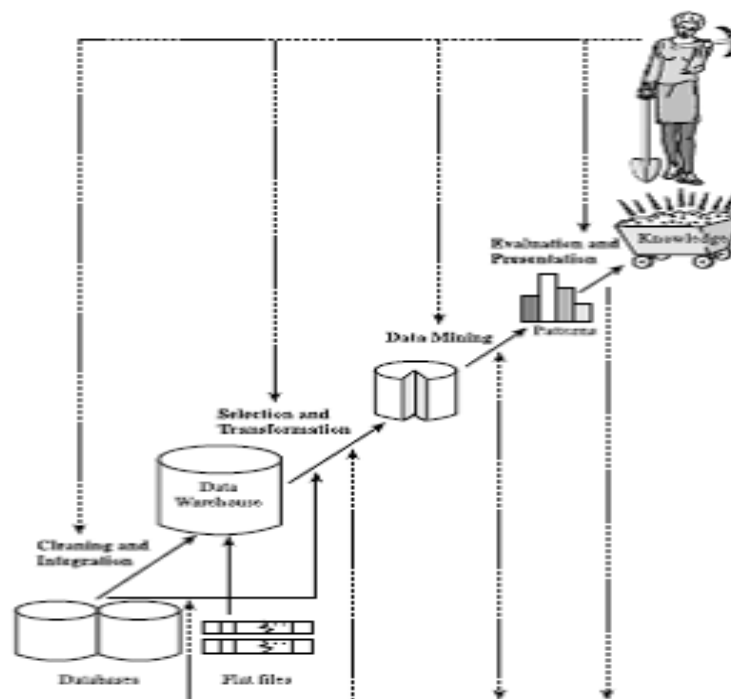
DATA MINING

What Is Data Mining?

- Data mining (knowledge discovery from data) Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
- Also called as Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.

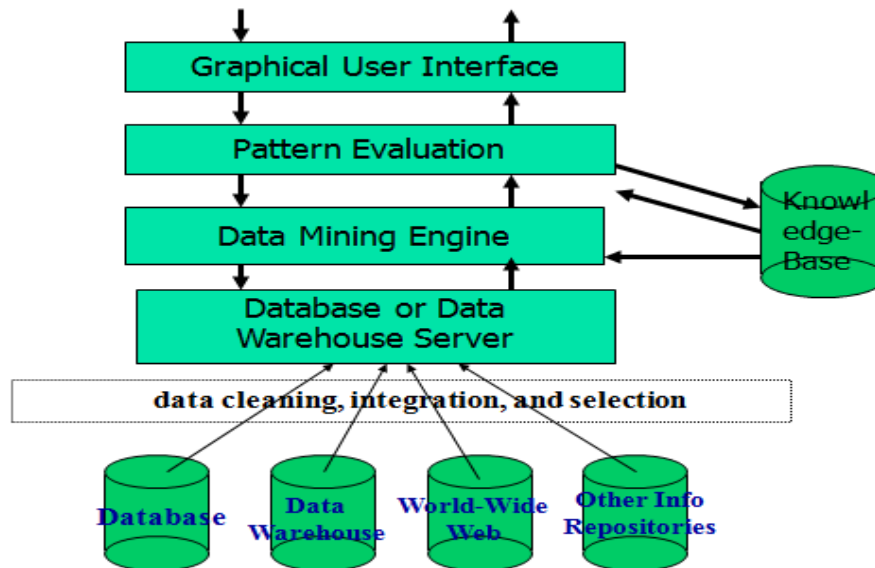
Steps in Knowledge Discovery

1. Data cleaning (to remove noise and inconsistent data)
2. Data integration (where multiple data sources may be combined)
3. Data selection (where data relevant to the analysis task are retrieved from the database)
4. Data transformation (where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations, for instance)
5. Data mining (an essential process where intelligent methods are applied in order to extract data patterns)
6. Pattern evaluation (to identify the truly interesting patterns representing knowledge based on some interestingness measures;
7. Knowledge presentation (where visualization and knowledge representation techniques are used to present the mined knowledge to the user)



Q) Explain The architecture of a Typical Data mining System

Architecture: Typical Data Mining System



Components of a data mining System

- **Knowledge base:** This is the domain knowledge that is used to **guide the search or evaluate the interestingness of resulting patterns**. Such knowledge can include concept hierarchies, used to organize attributes or attribute values into different levels of abstraction. Knowledge such as user beliefs, which can be used to assess a pattern's interestingness based on its unexpectedness, may also be included
- **Data mining engine:** This is essential to the data mining system and ideally consists of a set of **functional modules for tasks such as characterization, association and correlation analysis, classification, prediction, cluster analysis, outlier analysis, and evolution analysis**.
- **Pattern evaluation module:** This component typically employs interestingness measures and interacts with the data mining modules so as to *focus* the search toward interesting patterns. It may use **interestingness thresholds** to filter out discovered patterns. For efficient data mining, it is highly recommended to push the evaluation of pattern interestingness as deep as possible into the mining process so as to confine the search to only the interesting patterns.
- **User interface:** This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to **help focus the search, and investigating data mining based on the intermediate data mining results**. In addition, this component allows the user to browse database and data warehouse schemas or data structures, evaluate mined patterns, and visualize the patterns in different forms.

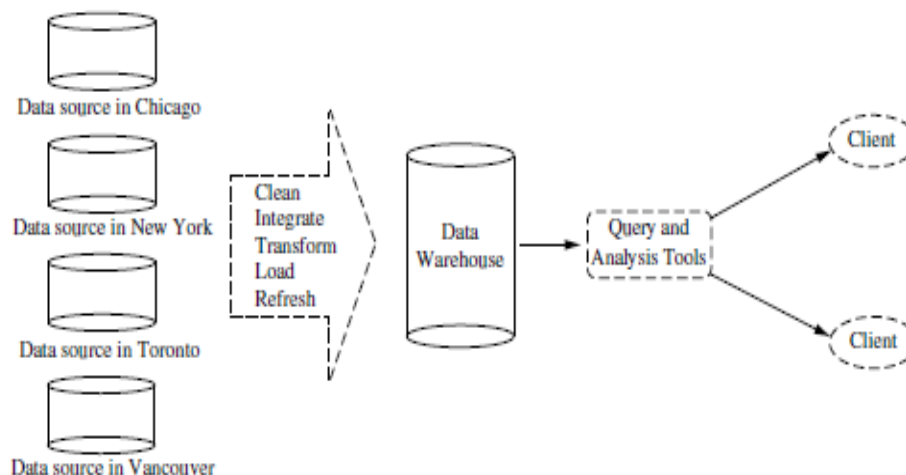
Lecture-2

On What Kind Data Can we perform Data Mining?

Data Mining can be performed on various types of data. These are Broadly divided into two types. Database Oriented data sets and advanced Data sets.

■ Database-oriented data sets and applications

1. Relational database, data warehouse, transactional database-RDBMS,DBMS



■ Advanced data sets and advanced applications

- Data streams and sensor data- huge or possibly infinite volume, dynamically changing, flowing in and out in a fixed order, allowing only one or a small number of scans, and demanding fast (often real-time) response time. Typical examples of data streams include various kinds of scientific and engineering data, time-series data, and data produced in other dynamic environments, such as power supply, network traffic, stock exchange, telecommunications, Web click streams, video surveillance, and weather or environment monitoring.
- Temporal data- (incl. bio-sequences) A temporal database typically stores relational data that include time-related attributes. These attributes may involve several timestamps, each having different semantics.
- A sequence database stores sequences of ordered events, with or without a concrete notion of time. Examples include customer shopping sequences, Web click streams, and biological sequences.
- A time-series database -stores sequences of values or events obtained over repeated measurements of time (e.g., hourly, daily, weekly). Examples include data collected from the stock exchange, inventory control, and the observation of natural phenomena (like temperature and wind).

- Object-relational databases- This model extends the relational model by providing a rich data type for handling complex objects and object orientation.
- Heterogeneous databases and legacy databases -A heterogeneous database consists of a set of interconnected, autonomous component. A legacy database is a group of *heterogeneous databases* that combines different kinds of data systems, such as relational or object-oriented databases, hierarchical databases, network databases, spreadsheets, multimedia databases, or file systems. The heterogeneous databases in a legacy database may be connected by inter-computer networks
- Spatial data and spatiotemporal data :Spatial databases contain spatial-related information. Examples include geographic (map) databases, very large-scale integration (VLSI) or computer-aided design databases, and medical and satellite image databases. A spatial database that stores spatial objects that change with time is called a spatiotemporal database, from which interesting information can be mined
- Multimedia database
- Text databases & WWW

Lecture-3

What are the various Data Mining Functionalities/Task and What Kinds of Patterns Can Be Mined?

Data mining tasks can be classified into two categories: descriptive and predictive.

1. Descriptive mining tasks characterize the general properties of the data in the database.
2. Predictive mining tasks perform inference on the current data in order to make predictions.

i. Concept/Class Description: Characterization and Discrimination

Data can be associated with classes or concepts.

For example, in the *AllElectronics* store, classes of items for sale include *computers* and *printers*, and concepts of customers include *bigSpenders* and *budgetSpenders*. Concept of students can be regular or irregular.

Such descriptions of a class or a concept are called class/concept descriptions. These descriptions can be derived via

- (1) *data characterization*, by summarizing the data of the class under study and putting it in a target class.
- (2) *data discrimination*, by comparison of the target class with one or a set of comparative classes
- (3) both data characterization and discrimination.

ii. Mining Frequent Patterns, Associations, and Correlations

- Frequent patterns, as the name suggests, are patterns that occur frequently in data. There are many kinds of frequent patterns, including itemsets, subsequences, and substructures.

- A *frequent itemset* typically refers to a set of items that frequently appear together in a transactional data set, such as milk and bread

Association analysis :It is the analysis based on association Rules.

It can be single dimensional or multidimensional.

An example of such a rule, mined from the *AllElectronics* transactional database, is $buys(X, \text{"computer"}) \rightarrow buys(X, \text{"software"})$ [support = 1%; confidence = 50%] where X is a variable representing a customer. A confidence, or certainty, of 50% means that if a customer buys a computer, there is a 50% chance that she will buy software as well. A 1% support means that 1% of all of the transactions under analysis showed that computer and software were purchased together.

Q) Explain this rule:-

A data mining system may find association rules like

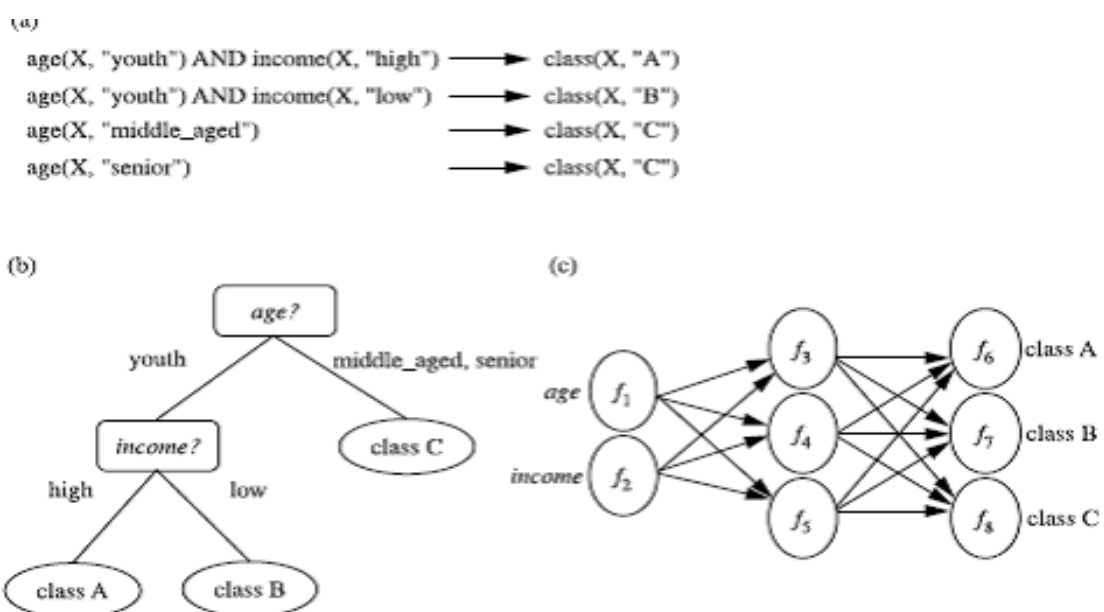
$age(X, \text{"20:::29"}) \wedge income(X, \text{"20K:::29K"}) \rightarrow buys(X, \text{"CD player"})$ [support = 2%, confidence = 60%]

Ans) The rule indicates that of the customers under study, 2% are 20 to 29 years of age with an income of 20,000 to 29,000 and have purchased a CD player at *AllElectronics*. There is a 60% probability that a customer in this age and income group will purchase a CD player. Note that this is an association between more than one attribute, or predicate (i.e., *age*, *income*, and *buys*). So it is multidimensional.

iii. Classification and Prediction

Classification is the process of finding a model (or function) that describes and distinguishes data classes or concepts, for the purpose of being able to use the model to predict the class of objects. The derived model is based on the analysis of a set of training data (i.e., data objects whose class label is known).

Classification Models $\rightarrow$ If then else, decision Tree, Neural Networks.



iv.Cluster Analysis

The objects are clustered or grouped based on the principle of *maximizing the intraclass similarity and minimizing the interclass similarity*. That is, clusters of objects are formed so that objects within a cluster have high similarity in comparison to one another, but are very dissimilar to objects in other clusters. Each cluster that is formed can be viewed as a class of objects. Examples: Group of customers in a city .

v.Outlier Analysis

A database may contain data objects that do not comply with the general behavior or model of the data. These data objects are outliers. Most data mining methods discard outliers as noise exceptions. However, in some applications such as fraud detection, the rare events can be more interesting than the more regularly occurring ones. The analysis of outlier data is referred to as outlier mining.

Lecture-4

Data Mining Task Primitives

A data mining query is defined in terms of data mining task primitives. These primitives allow the user to *interactively* communicate with the data mining system.

Q)List and describe the five *primitives* for specifying a data mining task.

The data mining primitives are:-

1. The set of *task-relevant data* to be mined-This includes the database attributes or data warehouse dimensions of interest (referred to as the *relevant attributes or dimensions*)
4. The *kind of knowledge* to be mined: This specifies the *data mining functions* to be performed, such as characterization, discrimination, association or correlation analysis, classification, prediction, clustering, outlier analysis, or evolution analysis.
5. The *background knowledge* to be used in the discovery process.-use of concept hierarchy.
6. The *interestingness measures and thresholds* for pattern evaluation-use of confidence and support.
7. The expected *representation for visualizing* the discovered patterns: This refers to the form in which discovered patterns are to be displayed, which may include rules, tables,charts, graphs, decision trees, and cubes.

Example:-

- (1) use database AllElectronics db =====> task relevant
- (2) use hierarchy location hierarchy for T.branch, age hierarchy for C.age =====> background knowledge.
- (3) mine classification as promising customers=====> kind of knowledge.
- (4) in relevance to C.age, C.income, I.type, I.placemade, T.branch

(5) from customer C, item I, transaction

T

(6) where I.item ID = T.item ID and C.cust ID = T.cust ID and C.income $\geq$ 40,000 and I.price $\leq$ 100

(7) group by T.cust ID

(8) having sum(I.price) $\leq$ 1,000

(9) display as rules =====> representation for visualization.

Classification of Data Mining Systems

- Classification according to **the kinds of databases mined**: A data mining system can be classified according to the kinds of databases mined. Database systems can be classified according to different criteria (such as data models, or the types of data or applications involved), each of which may require its own data mining technique. Data mining systems can therefore be classified accordingly.
- Classification according to **the kinds of knowledge mined**: Data mining systems can be categorized according to the kinds of knowledge they mine, that is, based on data mining functionalities, such as characterization, discrimination, association and correlation analysis, classification, prediction, clustering, outlier analysis, and evolution analysis.
- Classification according to **the kinds of techniques utilized**: Data mining systems can be categorized according to the underlying data mining techniques employed. These techniques can be described according to the degree of user interaction involved.
- Classification according to **the applications adapted**: Data mining systems can also be categorized according to the applications they adapt. For example, data mining systems may be tailored specifically for finance, telecommunications, DNA, stock markets, e-mail, and so on.

Integration of a Data Mining System with a Database or Data Warehouse System

Q) How to integrate or couple the DM system with a database (DB) system and/or a data warehouse (DW) system.

- **No coupling**: *No coupling* means that a DM system will not utilize any function of a DB or DW system.
- **Loose coupling**: *Loose coupling* means that a DM system will use some facilities of a DB or DW system, fetching data from a data repository managed by these systems, performing data mining, and then storing the mining results either in a file or in a designated place in a database or data warehouse. Loose coupling is better than no coupling because it can fetch any portion of data stored in databases or data warehouses by using query processing,
- **Semitight coupling**: *Semitight coupling* means that besides linking a DM system to a DB/DW system, efficient implementations of a few essential data mining primitives.
- **Tight coupling**: *Tight coupling* means that a DM system is smoothly integrated into the DB/DW system. This approach is highly desirable because it facilitates efficient implementations of data mining functions, high system performance, and an integrated information processing environment.

Another Probable Question

Describe the differences between the following approaches for the integration of a data mining system with a database or data warehouse system: *no coupling*, *loose coupling*, *semitight coupling*, and *tight coupling*. State which approach you think is the most popular, and why?

What are the Major Issues in Data Mining?

Mining methodology and user interaction issues:

- Mining different kinds of knowledge in databases-Because different users can be interested in different kinds of knowledge, data mining should cover a wide spectrum of data analysis and knowledge discovery tasks. These tasks may use the same database in different ways and require the development of numerous data mining techniques.
- Interactive mining of knowledge at multiple levels of abstraction: Because it is difficult to know exactly what can be discovered within a database, the data mining process should be *interactive*. Interactive mining allows users to focus the search for patterns, providing and refining data mining requests based on returned results.
- Incorporation of background knowledge: Background knowledge, or information regarding the domain under study, may be used to guide the discovery process and allow discovered patterns to be expressed in concise terms and at different levels of abstraction.
- Handling noisy or incomplete data: The data stored in a database may reflect noise, exceptional cases, or incomplete data objects. When mining data regularities, these objects may confuse the process, causing the knowledge model constructed to overfit the data.

Performance issues: These include efficiency, scalability, and parallelization of data mining algorithms.

- Efficiency and scalability of data mining algorithms: To effectively extract information from a huge amount of data in databases, data mining algorithms must be efficient and scalable.
- Parallel, distributed, and incremental mining algorithms: The huge size of many databases, the wide distribution of data, and the computational complexity of some data mining methods are factors motivating the development of parallel and distributed data mining algorithms.

Issues relating to the diversity of database types:

- Handling of relational and complex types of data: Because relational databases and data warehouses are widely used, the development of efficient and effective data mining systems for such data is important.
- Mining information from heterogeneous databases and global information systems: Local- and wide-area computer networks (such as the Internet) connect many sources of data, forming huge, distributed, and heterogeneous databases.

Data Preprocessing

➤ **What is dirty Data?**

Data in the real world is dirty which means that they are incomplete ,noisy and inconsistent.

1. **INCOMPLETE**: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - a) e.g., occupation=""
 - b) Incomplete data may come from "Not applicable" data value when collected
 - c) Human/hardware/software problems
2. **NOISY**: containing errors or outliers e.g., Salary="-10"
They arise due to Faulty data collection instruments, Human or computer error at data entry Errors in data transmission
3. **INCONSISTENT**: containing discrepancies in codes or names
 - a) e.g., Age="42" Birthday="03/07/1997"
 - b) e.g., Was rating "1,2,3", now rating "A, B, C"
 - c) e.g., discrepancy between duplicate records

➤ **Why Is Data Preprocessing Important?**

No quality data leads to no quality mining results. Quality decisions must be based on quality data.

e.g., duplicate or missing data may cause incorrect or even misleading statistics.
Data warehouse needs consistent integration of quality data.

➤ **What are the Major Tasks in Data Preprocessing?**

1. Data cleaning-Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
2. Data integration-Integration of multiple databases, data cubes, or files
3. Data transformation-Normalization and aggregation
4. Data reduction-Obtains reduced representation in volume but produces the same or similar analytical results
5. Data discretization -Part of data reduction but with particular importance, especially for numerical data

In summary, real-world data tend to be dirty, incomplete, and inconsistent. Data preprocessing techniques can improve the quality of the data, thereby helping to improve the accuracy and efficiency of the subsequent mining process. Data preprocessing is an important step in the knowledge discovery process, because quality decisions must be based on quality data. Detecting data anomalies, rectifying them early, and reducing the data to be analyzed can lead to huge payoffs for decision making.

Descriptive Data Summarization

Defination:-Descriptive data summarization techniques can be used to identify the typical properties of your data and highlight which data values should be treated as noise or outliers.

It is of two types:-

1. central tendency include *mean, median, mode, and midrange*
2. Dispersion of the data. include *quartiles, interquartile range (IQR), and variance.*

Measuring the Central Tendency

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Mean:-

Its of two types :-

- Weighted arithmetic mean: when along with values weight are also attached
- Trimmed mean: chopping extreme values and concentrating on the mean values only.
- Mean is an algebraic measure.

Median: Middle value if odd number of values or average of the middle two values otherwise

Estimated by interpolation

- A holistic measure is a measure that must be computed on the entire data set as a whole. It cannot be computed by partitioning the given data into subsets and merging the values obtained for the measure in each subset. The median is an example of a holistic measure. Holistic measures are much more expensive to compute than distributive measures

$$median = L_1 + \left(\frac{n/2 - (\sum f)l}{f_{median}} \right) c$$

•

Mode-Value that occurs most frequently in the data

- Its types are Unimodal, bimodal, trimodal

- Empirical formula: $\text{mean-mode} = 3 * (\text{mean} - \text{median})$

Measuring the Dispersion of Data

■ Quartiles, outliers and boxplots

1. Quartiles: Q_1 (25th percentile), Q_3 (75th percentile)
2. Inter-quartile range: $IQR = Q_3 - Q_1$
3. Five number summary: min, Q_1 , M, Q_3 , max
4. Boxplot: ends of the box are the quartiles, median is marked, whiskers, and plot outlier individually
5. Outlier: usually, a value higher/lower than $1.5 \times IQR$

■ Variance and standard deviation (*sample: s , population: σ*)

1. Variance: (algebraic, scalable computation)

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^n x_i^2 - \mu^2$$

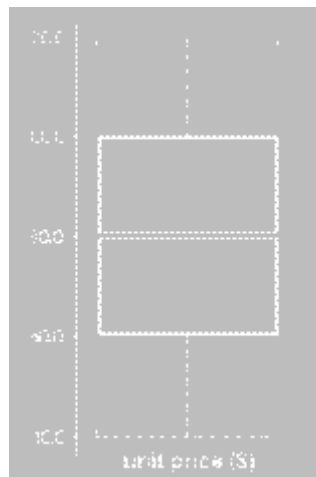
- Standard deviation s (*or* σ) is the square root of variance s^2 (*or* σ^2)

Box plot Analysis

- Five-number summary of a distribution: Minimum, Q_1 , M, Q_3 , Maximum

■ Box plot

1. Data is represented with a box
2. The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
3. The median is marked by a line within the box
4. Whiskers: two lines outside the box extend to Minimum and Maximum



Numerical example:-

1. Suppose that the data for analysis includes the attribute *age*. The *age* values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. (50 points)

- (a) What is the *mean* of the data? What is the *median*? What is the *standard deviation*?

$$\text{Mean} = (13+15+16+16+19+20+20+21+22+22+25+25+25+25+30+33+33+35+35+35+35+36+40+45+46+52+70) / 27 = 809/27 = 29.96$$

$$\text{Median} = \text{middle value} = 25$$

Standard deviation

Deviations from mean = {-16.96, -14.96, -13.96, -13.96, -10.96, -9.96, -9.96, -8.96, -7.96, -7.96, -4.96, -4.96, -4.96, -4.96, 0.04, 3.04, 3.04, 5.04, 5.04, 5.04, 5.04, 6.04, 10.04, 15.04, 16.04, 22.04, 40.04}

Squared deviations from mean = { 287.6416, 223.8016, 194.8816, 194.8816, 120.1216, 99.2016, 99.2016, 80.2816, 63.3616, 63.3616, 24.6016, 24.6016, 24.6016, 24.6016, 0.0016, 9.2416, 9.2416, 25.4016, 25.4016, 25.4016, 25.4016, 36.4816, 100.8016, 226.2016, 257.2816, 485.7616, 1603.2016}

Sum of squared deviations = 287.6416, 223.8016, 194.8816, 194.8816, 120.1216, 99.2016, 99.2016, 80.2816, 63.3616, 63.3616, 24.6016, 24.6016, 24.6016, 24.6016, 0.0016, 9.2416, 9.2416, 25.4016, 25.4016, 25.4016, 25.4016, 36.4816, 100.8016, 226.2016, 257.2816, 485.7616, 1603.2016 = 4354.9632

$$\text{Mean of squared deviations} = 4354.9632 / 27 = 161.2949$$

$$\text{Standard Deviation} = \sqrt{161.2949} = 12.7$$

What is the *mode* of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.).

Mode = most frequent number = {25, 35}

Modality = bimodal

(b) What is the *midrange* of the data?

$$\text{Midrange} = (13+70)/2 = 41.5$$

(c) Can you find (roughly) the 1st quartile (Q1) and the third quartile (Q3) of the data?

$$\text{Q1} = \text{median of lower half} = 20 \quad \text{Q3} = \text{median of upper half} = 35$$

(d) Give the *five-number summary* of the data.

$$\text{min, Q1, median, Q3, max} = 13, 20, 25, 35, 70$$

(e) If you plot a *boxplot* of this data, what will be the box length (in actual number)? What min and max value would the whisker extend to? Is there any outlier (i.e., elements beyond the extreme low and high)?

$$\text{Length} = \text{IQR} = \text{Q3} - \text{Q1} = 35 - 20 = 15$$

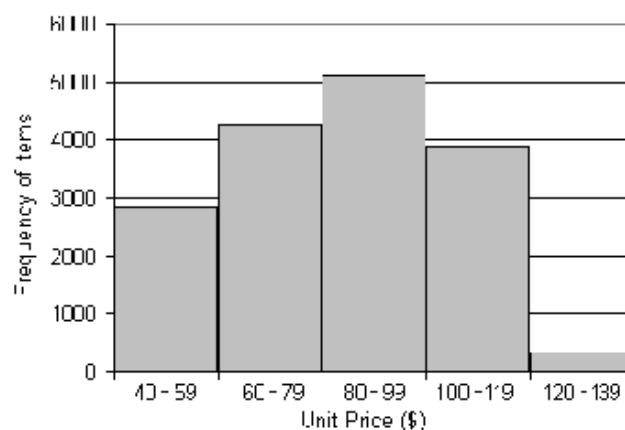
Min whisker = $\text{MAX}(Q1 - 1.5 * \text{IRQ}, \text{min}) = \text{MAX}(20 - 1.5 * 15, 13) = \text{MAX}(-2.5, 13) = 13$

Max whisker = $\text{MIN}(Q3 + 1.5 * \text{IRQ}, \text{max}) = \text{MIN}(35 + 1.5 * 15, 70) = \text{MIN}(67.5, 70) = 67.5$

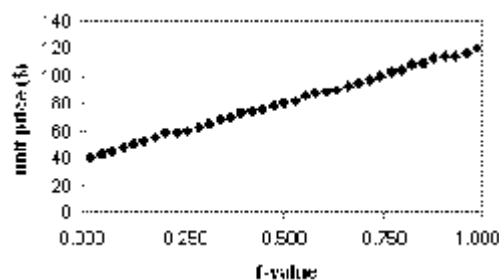
Outliers = anything higher than 67.5 or lower than 13 = {70}

Graphic Displays of Basic Descriptive Data Summaries

Histogram: Plotting histograms, or frequency histograms, is a graphical method for summarizing the distribution of a given attribute. A histogram for an attribute A partitions the data distribution of A into disjoint subsets, or *buckets*. Typically, the width of each bucket is uniform. Each bucket is represented by a rectangle whose height is equal to the count or relative frequency of the values at the bucket.



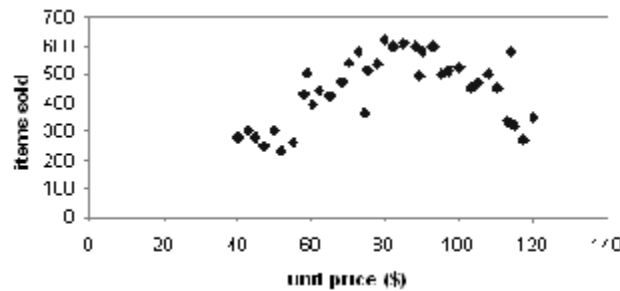
Quantile plot: A quantile plot is a simple and effective way to have a first look at a univariate data distribution. First, it displays all of the data for the given attribute (allowing the user to assess both the overall behavior and unusual occurrences). Second, it plots quantile information.



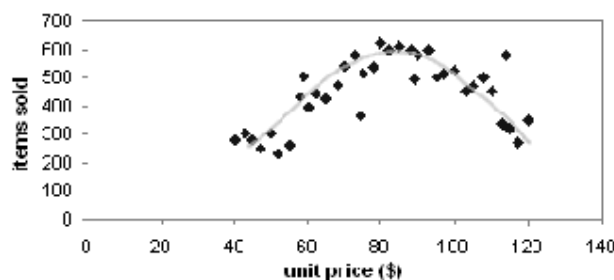
Quantile-quantile (q-q) plot: A quantile-quantile plot, or q-q plot, graphs the quantiles of one univariate

distribution against the corresponding quantiles of another. It is a powerful visualization tool in that it allows the user to view whether there is a shift in going from one distribution to another.

Scatter plot: A scatter plot is one of the most effective graphical methods for determining if there appears to be a relationship, pattern, or trend between two numerical attribute.



Loess (local regression) curve: A loess curve is another important exploratory graphic aid that adds a smooth curve to a scatter plot in order to provide better perception of the pattern



of dependence.

➤ What are the Major Tasks in Data Preprocessing?

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Data Cleaning

Data cleaning (or *data cleansing*) routines attempt to fill in missing values, smooth out noise while identifying outliers, and correct inconsistencies in the data.

Question: In real-world data, tuples with *missing values* for some attributes are a common occurrence. Describe various methods for handling this problem.

Answer: The various methods for handling the problem of missing values in data tuples include:

(a) Ignoring the tuple: This is usually done when the value is missing. This method is not very effective unless the tuple contains several attributes with missing values. It is especially poor when the percentage of missing values per attribute varies considerably.

(b) Manually filling in the missing value: In general, this approach is time-consuming and may not be a reasonable task for large data sets with many missing values, especially when the value to be filled in is not easily determined.

(c) Using a global constant to fill in the missing value: Replace all missing attribute values by the same constant, such as a label like "Unknown." Hence, although this method is simple, it is not recommended.

(d) Using the attribute mean for quantitative (numeric) values or attribute mode for categorical (nominal) values: For example, suppose that the average income of *AllElectronics* customers is \$28,000. Use this value to replace any missing values for *income*.

(e) Using the most probable value to fill in the missing value

Q:How do we remove Noisy Data?

Noise is random error or variance in a measured variable. It is a disturbance. The various reasons for noise are:

- Incorrect attribute values may due to
 1. faulty data collection instruments
 2. data entry problems
 3. data transmission problems
 4. technology limitation
 5. inconsistency in naming convention
- Other data problems which requires data cleaning
 1. duplicate records
 2. incomplete data
 3. inconsistent data

The various methods of data smoothing and removing noise are:-

1. Binning

- a) first sort data and partition into (equal-frequency/equal -width) bins
- b) then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.

Q: Suppose a group of 12 sales price records has been sorted as follows:

5; 10; 11; 13; 15; 35; 50; 55; 72; 92; 204; 215:

Partition them into three bins by each of the following methods.

- (a) equal-frequency partitioning
- (b) equal-width partitioning

Answer:

(a) equal-frequency partitioning

bin 1 5,10,11,13

bin 2 15,35,50,55

bin 3 72,92,204,215

(b) equal-width partitioning

The width of each interval is $(215 - 5)/3 = 70$.

bin 1 5,10,11,13,15,35,50,55,72

bin 2 92

bin 3 204,215

2. **Regression** : Data can be smoothed by fitting the data to a function, such as with regression. *Linear regression* involves finding the “best” line to fit two attributes (or variables), so that one attribute can be used to predict the other.

3. **Clustering**-detect and remove outliers-Outliers may be detected by clustering, where similar values are organized into groups, or “clusters.” Intuitively, values that fall outside of the set of clusters may be considered

Q:Explain Data Cleaning as a Process

The first step in data cleaning as a process is *discrepancy detection*. Discrepancies can be caused by several factors, including poorly designed data entry forms that have many optional fields, human error in data entry, deliberate errors (e.g., respondents not wanting to divulge information about themselves), and data decay (e.g., outdated addresses).Discrepancies may also arise from inconsistent data representations and the inconsistent use of codes.

Discrepancy can be solved by the following methods:-

- Use metadata (e.g., domain, range, dependency, distribution)
- Check field overloading
- Check uniqueness rule, consecutive rule and null rule
- Use commercial tools:-

- a) Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
- b) Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

DATA INTEGRATION

Data integration, combines data from multiple sources into a coherent data store, as in data warehousing. These sources may include multiple databases, data cubes, or flat files.

Q:Discuss issues to consider during data integration.

Answer: Data integration involves combining data from multiple sources into a coherent data store. Issues that must be considered during such integration include:

1. Entity-Identification problems/ Schema integration: The **metadata** from the different data sources must be integrated in order to match up equivalent real-world entities.
2. Handling redundant data: Derived attributes may be redundant, and inconsistent attribute naming may also lead to redundancies in the resulting data set. Duplications at the tuple level may occur and thus need to be detected and resolved.
3. Detection and resolution of data value conflicts: Differences in representation, scaling, or encoding may cause the same real-world entity attribute values to differ in the data sources being integrated.

Redundancy is another important issue:

Some redundancies can be detected by correlation analysis. Given two attributes, such analysis can measure how strongly one attribute implies the other, based on the available data. For numerical attributes, we can evaluate the correlation between two attributes, A and B, by computing the correlation coefficient.

- Correlation coefficient (also called Pearson's product moment coefficient)

$$r_{A,B} = \frac{\sum (A - \bar{A})(B - \bar{B})}{(n-1)\sigma_A\sigma_B} = \frac{\sum (AB) - n\bar{A}\bar{B}}{(n-1)\sigma_A\sigma_B}$$

where n is the number of tuples, respective means of A and B, σ_A and σ_B are the respective standard deviation of A and B, and $\sum(AB)$ is the sum of the AB cross-product.

1. If $r_{A,B} > 0$, A and B are positively correlated (A's values increase as B's). The higher, the stronger correlation.
2. $r_{A,B} = 0$: independent; $r_{A,B} < 0$: negatively correlated

- χ^2 (chi-square) test

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

- The larger the χ^2 value, the more likely the variables are related

DATA TRANSFORMATION

In *data transformation*, the data are transformed or consolidated into forms appropriate for mining. Data transformation can involve the following:

1. Smoothing, which works to remove noise from the data. Such techniques include binning, regression, and clustering.

2. Aggregation, where summary or aggregation operations are applied to the data. For example, the daily sales data may be aggregated so as to compute monthly and annual total amounts. This step is typically used in constructing a data cube for analysis of the data at multiple granularities.
3. Generalization of the data, where low-level or “primitive” (raw) data are replaced by higher-level concepts through the use of concept hierarchies. For example, categorical attributes, like *street*, can be generalized to higher-level concepts, like *city* or *country*. Similarly, values for numerical attributes, like *age*, may be mapped to higher-level concepts, like *youth*, *middle-aged*, and *senior*.
4. Normalization, where the attribute data are scaled so as to fall within a small specified range, such as 1:0 to 1:0, or 0:0 to 1:0.
5. Attribute construction (or *feature construction*), where new attributes are constructed and added from the given set of attributes to help the mining process.

Data Transformation: Normalization techniques

- Min-max normalization: to $[new_min_A, new_max_A]$

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Z-score normalization (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma}$$

- Normalization by decimal scaling

$$v' = \frac{v}{10^j}$$

DATA REDUCTION STRATEGIES

Data reduction techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data.

Strategies for data reduction include the following:

1. Data cube aggregation, where aggregation operations are applied to the data in the construction of a data cube.

2. Attribute subset selection, where irrelevant, weakly relevant, or redundant attributes or dimensions may be detected and removed. The following are the methods:-

i). Stepwise forward selection: The procedure starts with an empty set of attributes as the reduced set. The best of the original attributes is determined and added to the reduced set. At each subsequent iteration or step, the best of the remaining original attributes is added to the set.

ii). Stepwise backward elimination: The procedure starts with the full set of attributes. At each step, it removes the worst attribute remaining in the set.

iii). *Combination of forward selection and backward elimination*: The stepwise forward selection and backward elimination methods can be combined so that, at each step, the procedure selects the best attribute and removes the worst from among the remaining attributes.

3. Dimensionality reduction, where encoding mechanisms are used to reduce the data set size. In *dimensionality reduction*, data encoding or transformations are applied so as to obtain a reduced or “compressed” representation of the original data. If the original data can be *reconstructed* from the compressed data without any loss of information, the data reduction is called lossless. If, instead, we can reconstruct only an approximation of the original data, then the data reduction is called lossy. Two popular and effective methods of lossy dimensionality reduction: *wavelet transforms* and *principal components analysis*

4. Numerosity reduction, where the data are replaced or estimated by alternative, smaller data representations such as parametric models (which need store only the model parameters instead of the actual data) or nonparametric methods such as clustering, sampling, and the use of histograms.

Sampling

Sampling can be used as a data reduction technique because it allows a large data set to be represented by a much smaller random sample (or subset) of the data. Suppose that a large data set, D , contains N tuples

Simple random sample without replacement (SRSWOR) of size s : This is created by drawing s of the N tuples from D ($s < N$), where the probability of drawing any tuple in D is $1/N$, that is, all tuples are equally likely to be sampled.

Simple random sample with replacement (SRSWR) of size s : This is similar to SRSWOR, except that each time a tuple is drawn from D , it is recorded and then *replaced*. That is, after a tuple is drawn, it is placed back in D so that it may be drawn again

5. Discretization and concept hierarchy generation, where raw data values for attributes are replaced by ranges or higher conceptual levels Discretization and concept hierarchy generation are powerful tools for data mining, in that they allow the mining of data at multiple levels of abstraction. Discretization techniques can be categorized based on how the discretization is performed, such as whether it uses class information or which direction it proceeds (i.e., top-down vs. bottom-up)